

# Julien Pontoire

## Data Science & Artificial Intelligence Engineer

UTC 2026 Engineer · Final-year internship at Airbus Defence and Space · AI Agents in industrial context

Available from September 2026

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## Experience

### AI Engineering Internship — Airbus Defence and Space

February 2026 – July 2026

*Workflow automation · LLM Agents · Software engineering*

France

- **Open-source AI tools:** deployment and support for engineers' adoption in an industrial environment (OpenCode, OpenClaw)
- **Prompting language:** development to automate software engineering tasks
- **LLM pipeline:** development of an automatic defect correction pipeline from Jira tickets
- **Prompt testing:** set up a test suite with Promptfoo to evaluate and validate developed prompts
- **First place at my unit's hackathon (team project):** proof of concept of embedded AI (Edge AI), with an LLM running locally in the browser

### Assistant Engineer Internship — Sciences Po Medialab

September 2024 – February 2025

*Data collection and processing*

Paris, France

- Improved and developed data collection and processing tools for the laboratory
- **Rust:** implemented stemming algorithms (Porter, Carry) in a library
- Built data collection pipelines (Reddit, press websites) for scientific research
- Developed a bookmarklet to aggregate results from multiple search engines

## Projects

### AI Personal Training Assistant (RAG + Fine-tuning + Agent)

August 2026

- LangGraph agent orchestrating RAG (ChromaDB) and a fine-tuned model (Qwen 2.5, LoRA/QLoRA) to generate personalized training programs, with validated structured output (Pydantic) and retry logic
- REST API (FastAPI) and interface (Streamlit), deployed on Google Cloud Run with a multi-service architecture (Secret Manager, user API key management)

### VR Narrator with Generative AI

February – July 2025

- Research project: developed an AI-powered narrator in a mixed reality scene for the Memorial of Internment and Deportation in Compiègne

### Student Path Prediction

February – July 2025

- Analyzed a higher education dataset to predict students' academic outcomes
- Applied supervised methods (logistic regression, random forests) after PCA and unsupervised exploration

## Education

### Université de Technologie de Compiègne (UTC)

2021 – 2026

*Engineering degree in Computer Science — GPA: 4.63/5*

Compiègne, France

- **Commitment:** Elected student representative on the Academic and University Life Council (2023 – 2025)

### Université du Québec à Chicoutimi (UQAC)

August – December 2025

*International exchange semester*

Chicoutimi, Canada

- Business Intelligence, video games, parallel architecture, metaheuristics

### Baccalauréat Général

July 2021

*Mention Très Bien — Specialization in Mathematics and Engineering Sciences*

Lycée Boucher de Perthes, Abbeville

## Technical Skills

**AI & Data Science:** LLM Agents, Prompting, MCP, Promptfoo, Ollama, Python, NumPy, Pandas, Scikit-learn, PySpark, BeautifulSoup, PyTorch, LangChain, LangGraph, RAG, Fine-tuning (LoRA/QLoRA), Hugging Face (Transformers, PEFT, TRL), ChromaDB, Qdrant, FastAPI, Pydantic

**Software development:** Rust, Java, C, JavaScript (VueJS, React, SpringBoot), Bash

**Data & BI:** PostgreSQL, Snowflake, phpMyAdmin, Power BI, ETL, Gephi

**DevOps & Tools:** Git, GitLab / CI-CD, Docker, Linux, VS Code, JetBrains, Google Cloud Platform (Cloud Run, Artifact Registry, Secret Manager)

## Languages

**French:** Native speaker   **English:** C1 — TOEIC 980/990   **Japanese:** A1   **Spanish:** A2